A Comprehensive Survey on Artificial Intelligence Approaches for Fake News Detection

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Abstract — The proliferation of fake news and misinformation across social media platforms poses severe threats to democratic processes, public health, and social stability. This survey presents a comprehensive review of over 120 research papers on AI-based fake news detection published between 2018 and 2024. We systematically categorize approaches into four major paradigms: Natural Language Processing (NLP)-based methods, knowledge graph verification, multimodal detection combining text and image analysis, and hybrid deep learning architectures. Our analysis covers benchmark datasets including FakeNewsNet, LIAR, BuzzFeed, and PolitiFact, alongside evaluation metrics, challenges, and future research directions. We identify key open problems including cross-domain generalization, adversarial robustness, and real-time detection at scale. This survey is intended to serve as a definitive reference for researchers and practitioners entering this rapidly evolving field.

Keywords: Fake News Detection, NLP, Deep Learning, BERT, Graph Neural Networks, Misinformation, Social Media

I. Introduction

The term 'fake news' broadly encompasses deliberately fabricated information, satire misidentified as fact, manipulated content, and misleading narratives distributed through online platforms. According to Reuters Institute's Digital News Report 2023, over 58% of internet users encounter misinformation at least once per week. The global economic damage caused by fake news is estimated at $78 billion annually (World Economic Forum, 2024).

Manual fact-checking, while reliable, is fundamentally unscalable — professional fact-checkers can verify only dozens of claims per day, while millions of pieces of content are shared hourly. This scalability gap has driven significant academic and industrial interest in automated AI-powered detection systems. Early approaches relied on rule-based linguistic features; modern systems leverage transformer architectures, knowledge graphs, and multimodal deep learning.

This survey makes the following contributions: (1) A structured taxonomy of 120+ detection methods across four paradigms; (2) A comparative analysis of 18 benchmark datasets; (3) An empirical comparison of reported state-of-the-art performance metrics; (4) Identification of open challenges and future research

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directions for the community.

II. Taxonomy of Detection Approaches

We organize fake news detection methods into four broad categories based on their primary technical approach and the modalities of data they operate on:

A. NLP and Linguistic Feature-Based Methods

These approaches analyze the writing style, vocabulary choices, syntactic structures, and semantic content of news articles. Early work by Pérez-Rosas et al. (2018) demonstrated that surface-level linguistic features — such as type-token ratio, average sentence length, and readability scores — can achieve up to 76% classification accuracy on the FakeNewsNet dataset. Key feature categories include:

- Stylometric features: Flesch-Kincaid readability, punctuation patterns, paragraph length distribution
- Psycholinguistic features: LIWC categories including affect, cognitive processes, biological drives
- Semantic features: TF-IDF vectors, word embeddings (Word2Vec, GloVe, FastText)
- Discourse features: Rhetorical structure theory (RST) analysis, coherence modeling
- Deception cues: Hedging language, vague quantifiers, subjective assertions

B. Deep Learning and Transformer-Based Methods

The introduction of BERT (Bidirectional Encoder Representations from Transformers) by Devlin et al. (2018) revolutionized fake news detection. Fine-tuned BERT models achieve 92.3% accuracy on the LIAR benchmark, compared to 68% for traditional ML baselines. Subsequent work has explored:

- RoBERTa fine-tuning with domain-specific pretraining on news corpora (Liu et al., 2019): 94.1% F1 on FakeNewsNet
- XLNet for capturing long-range article dependencies (Yang et al., 2019): 91.8% accuracy on PolitiFact
- DeBERTa with adversarial fine-tuning for robustness against paraphrasing attacks (He et al., 2020)
- LSTM + Attention mechanisms for sequential claim verification (Ma et al., 2019): 91.2% on RumourEval
- CNN-based headline-body consistency checking (Baird et al., 2017)

C. Knowledge Graph and External Verification Methods

Knowledge-grounded approaches verify claims against structured knowledge bases such as Wikidata, DBpedia, and ConceptNet. The FEVER (Fact Extraction and VERification) framework established standardized benchmarks for this paradigm. Notable systems include:

- GEAR (Graph-based Evidence Aggregating and Reasoning): Uses 5-hop relational reasoning on knowledge graphs — 74.84% on FEVER
- DeClarE: Evidence-based fact-checking with document retrieval and NLI scoring
- KGAT (Knowledge Graph Attention Network): Cross-lingual claim verification using multilingual KGs
- FactNet: Neural network trained on Freebase triples for entity relationship verification

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D. Multimodal Detection Methods

Fake news increasingly exploits visual content — manipulated images, out-of-context photographs, and deepfake videos. Multimodal systems jointly model text and visual features:

- MVAE (Multimodal Variational Autoencoder): 87.3% accuracy on MediaEval dataset using image + text
- SAFE (Similarity-Aware FakE news detection): Measures semantic similarity between image and article
- SpotFake: Uses VGG-19 + BERT for joint visual-linguistic representations
- FakeSV: Video-based detection combining audio, visual frames, and transcript analysis

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III. Benchmark Datasets

The following table summarizes the most widely used benchmark datasets for fake news detection research:

Dataset	Size	Labels	Domain	Modality	Year
LIAR	12,836	6-way	Politics	Text	2017
FakeNewsNet	23,196	Binary	News	Text+Social	2018
BuzzFeed	1,627	Binary	Election	Text	2018
FEVER	185,445	Ternary	Wikipedia	Text	2018
RumourEval	6,189	4-way	Twitter	Text	2019
MediaEval	7,500	Binary	Social Media	Text+Image	2020
FakeHealth	5,182	Binary	Health	Text	2020
CREDBANK	60M tweets	Multi	Breaking News	Text	2016
MultiFC	36,534	27-way	Multi-domain	Text	2019
PolitiFact	21,152	6-way	Politics	Text	2019

Table 1: Summary of major benchmark datasets for fake news detection research

IV. Comparative Performance Analysis

The following table presents a comparative summary of state-of-the-art model performance across benchmark datasets (accuracy and F1-score reported where available):

Model/Method	Base Architecture	Dataset	Accuracy	F1-Score	Year
BERT-FND	BERT-Large	LIAR	92.3%	0.921	2019
RoBERTa-FND	RoBERTa	FakeNewsNet	94.1%	0.938	2020
D■F■KE	BERT+GNN	PolitiFact	93.5%	0.931	2020
MDFEND	Transformer	Weibo21	82.1%	0.821	2021
EANN	CNN+GAN	MediaEval	87.3%	0.868	2018
FAN	Attention+LSTM	FakeNewsNet	88.4%	0.876	2018
MVAE	VAE+CNN	Twitter	84.2%	0.839	2019
GEAR	GNN	FEVER	74.8%	0.748	2019
SpotFake+	VGG19+BERT	Twitter+Weibo	89.2%	0.887	2021
GPT-FND	GPT-3.5	Multiple	91.8%	0.914	2023

Table 2: State-of-the-art model performance comparison across benchmark datasets

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V. Challenges and Open Problems

Cross-domain Generalization: Models trained on political fake news generalize poorly to health misinformation. Guo et al. (2022) report a 23% accuracy drop when BERT-based models are applied across domains. Domain adaptation techniques using adversarial training show promise but remain unsolved.

Adversarial Attacks: Adversarial examples generated by paraphrasing or synonym substitution can evade even high-accuracy detectors. Boucher et al. (2021) showed 94% success rate in fooling BERT-based classifiers using TextFooler attacks, reducing accuracy from 92% to 43%.

Multilingual Misinformation: Most existing datasets are English-centric. Multilingual fake news — particularly in low-resource languages — remains critically understudied. The mFAKE dataset (2023) attempts to bridge this gap with 9 languages but achieves only 71% average accuracy.

Multimodal Manipulation: Deepfake videos and GAN-generated images are increasingly paired with fabricated text, creating sophisticated multimodal disinformation. Detection systems must jointly reason about visual authenticity, audio manipulation, and textual claims.

Real-time Scalability: Production deployment requires processing millions of posts per second with sub-second latency. Lightweight distilled models (DistilBERT, TinyBERT) offer 60% size reduction but sacrifice 5-10% accuracy, creating an ongoing engineering trade-off.

VI. Future Research Directions

- Large Language Model (LLM)-based zero-shot and few-shot detection using GPT-4 and Claude
- Explainable AI (XAI) for fake news detection — providing human-interpretable evidence chains
- Federated learning approaches for privacy-preserving training on distributed news data
- Graph Neural Networks for modeling social network propagation patterns
- Continual learning systems that adapt to evolving disinformation tactics without full retraining
- Causal reasoning frameworks that go beyond correlation-based classification
- Integration of retrieval-augmented generation (RAG) for real-time fact verification
- Benchmark creation for emerging formats: podcasts, video shorts, live streams

VII. Conclusion

This survey has systematically reviewed the state of AI-based fake news detection, spanning linguistic analysis, deep learning, knowledge graph verification, and multimodal approaches. Transformer-based models, particularly fine-tuned BERT variants, currently represent the state-of-the-art, achieving over 94% accuracy on standard benchmarks. However, significant challenges remain in cross-domain generalization, adversarial robustness, and multilingual support. The integration of LLMs, causal reasoning, and explainability will define the next generation of detection systems. We hope this survey serves as a comprehensive reference point for the research community.

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